Hom Der

Deformation Theory Notes

Based on Har10

1 First order deformations

The main reference for Appendix is [**?**].

1.1 The T^i Functors

Let $A \to B$ be a homomorphism of rings and let M be a B-module. Choose the set of variables $x = \{x_i\}$ and a free R-module F such that

$$0 \longrightarrow I \longrightarrow R \longrightarrow B \longrightarrow 0, \qquad 0 \longrightarrow Q \longrightarrow F \longrightarrow I \longrightarrow 0$$

and let

$$F_0 = \langle j(a)b - j(b)a \mid a, b \in F \rangle_R \subseteq F.$$

Define a complex of B-modules, called the cotangent complex

$$L_2 \xrightarrow{d_2} L_1 \xrightarrow{d_1} L_0$$

as follows. Take $L_2 = Q/F_0$, which is a B-module. Indeed, if $x \in I$ and $a \in Q$. Write $x = j(x')$ for some $x' \in F$, then

$$xa = j(x')a \equiv j(a)x' = 0 \pmod{F_0}.$$

Take $L_1 = F \otimes_R B = F/IF$, and let $d_2 : L_2 \to L_1$ be the map induced from the inclusion $Q \to F$. Take $L_0 = \Omega_{R/A} \otimes_R B$, where $\Omega_{R/A}$ is the module of relative Kähler differentials. Let d_1 be the composition

$$L_1 = F \otimes_R B \longrightarrow \Omega_{R/A} \otimes_R B = L_0$$

induced by the universal derivation. It is clear that $d_1 d_2 = 0$, so we defined a complex of B-modules. Notice that L_0, L_1 are free B-modules, since R is a polynomial ring over A.

Indeed, the second fundamental exact sequence of Kähler differentials

$$I/I^2 \xrightarrow{d} \Omega_{R/A} \otimes_R B \to \Omega_{B/A} \to 0$$

gives the exact sequence

$$L_2 \xrightarrow{d_2} L_1 \xrightarrow{d_1} L_0 \to \Omega_{B/A} \to 0. \tag{10}$$

For any B-module M, define

$$T^i(B/A, M) := H^i(0_B(L_\bullet, M)),$$

which is independent of the choice of F and R.

Proof. First, we fix the polynomial ring R. If F and F' are two choices of free R-modules mapping onto I, then $F \oplus F'$ gives the third choice. Since F' is free, the map $j' : F' \to I$ factors through F, i.e., $j' = jp$ for some map $p : F' \to F$. Replacing each generator e' of $F' \subseteq F \oplus F'$ by $e' - p(e')$, we may assume that the map $F \oplus F' \to I$ is just $(j, 0)$. Then $\ker(j, 0) = Q \oplus F'$ and thus $(F \oplus F')_0 = F_0 + IF'$. Let $L'_\bullet$ be the complex obtained from $(F \oplus F', R)$, then

$$L'_2 = L_2 \oplus F'/IF', \quad L'_1 = L_1 \oplus (F' \otimes_R B), \quad L'_0 = L_0.$$

Hence, $L'_\bullet$ is obtained by the direct sum of $L_\bullet$ with the free acyclic complex $F' \otimes_R B \to F' \otimes_R B \to 0$, which induces same cohomology after taking 0 functor.

Second, it suffices to compare $R = A[x]$ with $R' = A[x, y]$. Again, the map $A[y] \to B$ can be factored through $A[x]$ by a homomorphism $p : A[y] \to A[x]$. Replacing each $y_i \in A[x, y]$ by $y_i - p(y_i)$, we may assume the ring homomorphism $A[x, y] \to B$ maps y_i to 0, then it has kernel $IR' + yR'$. Since T^i is independent of the choice of F, we may take $F' = F \otimes_R R'$ and G be a free R'-module on the index set of the y variables. Then we have the following diagram

$$\begin{array}{ccccccccc}
0 & \to & QR' & \to & F' & \to & IR' & \to & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \to & QR' + \langle y_i a - j(a)e_i \rangle & \to & F' \oplus G & \to & IR' + yR' & \to & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \to & P & \to & G & \to & J & \to & 0
\end{array}$$

and it is clear that

$$Q' = QR' + \langle y_i a - j(a)e_i \mid a \in F \rangle_{R'} + \langle y_i e_j - y_j e_i \rangle_{R'}.$$

Since $\langle y_i a - j(a)e_i \mid a \in F \rangle_{R'} + \langle y_i e_j - y_j e_i \rangle_{R'} \subseteq (F' \oplus G')_0$, we have $L'_2 = L_2$.

On the other hand, $L'_1 = L_1 \oplus (G' \otimes_{R'} B)$ and $L'_0 = L_0 \oplus (\Omega_{A[y]/A} \otimes B)$. Then $L'_\bullet$ is obtained by the direct sum of $L_\bullet$ with the free acyclic complex

$$G' \otimes_{R'} B \longrightarrow \Omega_{A[y]/A} \otimes B \longrightarrow 0, \qquad e_i \longrightarrow dy_i,$$

and thus induces same cohomology after taking 0 functor. □

Remark 1.1. The proof also shows that $L_\bullet$ is a well-defined element in the derived category of the category of B-modules.

Theorem 1.2. *Let $A \to B$ be a homomorphism of rings. Then $T^i(B/A, -)$ is a covariant, additive functor from Mod_B to itself. If*

$$0 \to M' \to M \to M'' \to 0$$

is a short exact sequence of B-modules, then there is a long exact sequence

$$\begin{aligned} 0 \longrightarrow T^0(B/A, M') &\longrightarrow T^0(B/A, M) \longrightarrow T^0(B/A, M'') \\ \longrightarrow T^1(B/A, M') &\longrightarrow T^1(B/A, M) \longrightarrow T^1(B/A, M'') \\ \longrightarrow T^2(B/A, M') &\longrightarrow T^2(B/A, M) \longrightarrow T^2(B/A, M''). \end{aligned}$$

Proof. By construction, $T^i(B/A, -)$ is a covariant additive functor. Since the terms L_1 and L_0 of the complex $L_\bullet$ are free, we get a sequence of complexes

$$0 \longrightarrow 0_B(L_\bullet, M') \longrightarrow 0_B(L_\bullet, M) \longrightarrow 0_B(L_\bullet, M'') \longrightarrow 0$$

that is exact except possibly for the map

$$0_B(L_2, M) \to 0_B(L_2, M'')$$

may not be surjective. This sequence of complexes gives the long exact sequence of cohomology above. □

Theorem 1.3. *Let $A \to B \to C$ be homomorphisms of rings, and let M be a C-module. Then there is a long exact sequence*

$$\begin{aligned} 0 \longrightarrow T^0(C/B, M) &\longrightarrow T^0(C/A, M) \longrightarrow T^0(B/A, M) \\ \longrightarrow T^1(C/B, M) &\longrightarrow T^1(C/A, M) \longrightarrow T^1(B/A, M) \\ \longrightarrow T^2(C/B, M) &\longrightarrow T^2(C/A, M) \longrightarrow T^2(B/A, M). \end{aligned}$$

Proof. Choose

$$0 \longrightarrow I \longrightarrow A[x] \longrightarrow B \longrightarrow 0, \qquad 0 \longrightarrow Q \longrightarrow F \longrightarrow I \longrightarrow 0$$

$$0 \longrightarrow J \longrightarrow B[y] \longrightarrow C \longrightarrow 0, \qquad 0 \longrightarrow P \longrightarrow G \longrightarrow J \longrightarrow 0$$

to calculate $T^i(B/A, M)$ and $T^i(C/B, M)$. Then $A[x, y] \to B[y] \to C$ is surjective, and denote its kernel by K. It induces an exact sequence

$$0 \longrightarrow I[y] \longrightarrow K \longrightarrow J \longrightarrow 0$$

of A-modules. Let F' and G' be free $A[x, y]$-modules on the same index sets as F and G respectively. Choose a lifting of the map $G \to J$ to a map $G' \to K$. From the map $F \to I$ we get the natural map $F' \to I[y]$. Hence, we get the surjection $F' \oplus G' \to K$, and let S be its kernel. Now, we can calculate $T^i(C/A, M)$ by

$$0 \to K \to A[x, y] \to C \to 0, \qquad 0 \to S \to F' \oplus G' \to K \to 0.$$

Now, we have the following diagram

$$\begin{array}{ccccccccc} 0 & \to & Q[y] & \to & F' & \to & I[y] & \to & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \to & S & \to & F' \oplus G' & \to & K & \to & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \to & P & \to & G & \to & J & \to & 0 \end{array}$$

which is exact for each horizontal sequence, but not exact for vertical sequence. It induces the maps of complexes

$$L_\bullet(B/A) \otimes_B C \to L_\bullet(C/A) \to L_\bullet(C/B).$$

Indeed, on the degree 0 level we have split exact sequence

$$\Omega_{A[x]/A} \otimes_{A[x]} C \to \Omega_{A[x,y]/A} \otimes_{A[x,y]} C \to \Omega_{B[y]/B} \otimes_{B[y]} C$$

given by $dx_i \mapsto dx_i$, $dx_i, dy_j \mapsto 0, dy_j$. On the degree 1 level we have split exact sequence

$$F \otimes_{A[x]} C \to (F' \oplus G') \otimes_{A[x,y]} C \to G \otimes_{B[y]} C$$

of free C-modules. On the degree 2 level we have

$$(Q/F_0) \otimes_B C \to S/(F' \oplus G')_0 \to P/G_0. \tag{11}$$

We claim that $S \to P$ is surjective. Indeed, if $p \in P$. Take $(f', g') \in F' \oplus G'$ mapping to p. Then the image of (f', g') in K is contained in $I[y]$. Since $F' \to I[y]$ is surjective, there exists $\tilde{f} \in F'$ having the same image in K as (f', g'). Hence, $(f' - \tilde{f}, g') \in S$ maps to p.

To complete the proof, we remain to show the exactness of (11) in the middle. Let $s = f + g$ be an element of $S \subseteq F' \oplus G'$ such that its image in P is contained in G_0. Then the image g of g' in G belongs to G_0, write $g = j(a)b - j(b)a$ for some $a, b \in G$. Lift a, b to elements a', b' in G', then $j(a')b' - j(b')a' \in S$. Let $g'' = g - j(a')b' + j(b')a'$, then $g'' \in \ker(G' \to G) = IG'$. Say $g'' = xh$ and $x = j(x')$ for some $x \in I$, $h \in G'$, and $x' \in F$. Then

$$xh = j(x')h \equiv j(h)x' \pmod{(F' + G')_0}.$$

Therefore,

$$s \equiv f' + g'' \equiv f' + j(h)x' \pmod{(F' + G')_0}$$

and $f' + j(h)x' \in F' \cap S = Q[y]$. □

Proposition 1.4. *For any morphism $A \to B$ and $M \in \mathsf{Mod}_B$,*

$$T^0(B/A, M) = 0_B(\Omega_{B/A}, M) = 0_A(B, M).$$

In particular,

$$T^0(B/A, B) = 0_B(\Omega_{B/A}, B) = T_{B/A}$$

is the tangent module of B over A.

Proof. Taking $0(-, M)$ to the exact sequence (10), we have the exact sequence

$$0 \to 0(\Omega_{B/A}, M) \to 0(L_0, M) \to 0(L_1, M).$$

□

Proposition 1.5. *If B is a polynomial ring over A, then $T^i(B/A, M) = 0$ for $i = 1, 2$ and for all M.*

Proof. We may take $R = B$ and $F = 0$ in our construction. Hence, the cotangent complex only has L_0 term. □

Proposition 1.6. *If $A \to B$ is a surjective ring homomorphism with kernel I, then*

$$T^0(B/A, M) = 0, \quad T^1(B/A, M) = 0_B(I/I^2, M)$$

for any $M \in \mathsf{Mod}_B$. In particular, $T^1(B/A, B) = 0_B(I/I^2, B)$ is the normal module $N_{B/A}$ of B in A.

Proof. In this case, we can take $R = A$, and thus $L_0 = 0$. Hence, $T^0 = 0$ for any module M. Furthermore, the exact sequence

$$0 \longrightarrow Q \longrightarrow F \xrightarrow{j} I \longrightarrow 0$$

tensored with B gives

$$Q \otimes_A B \longrightarrow F \otimes_A B \xrightarrow{j} I/I^2 \longrightarrow 0$$

and we have

$$L_2 \longrightarrow L_1 \longrightarrow I/I^2 \longrightarrow 0,$$

thus $T^1(B/A, M) = 0_B(I/I^2, M)$. □

Corollary 1.7. *Suppose that A is a local ring and $B = A/I$, where I is generated by a regular sequence $a_1, \ldots, a_r$. Then $T^2(B/A, M) = 0$ for all M.*

Proof. Since the Koszul complex of a regular sequence is exact, we conclude that $Q = F_0$. Hence, $L_2 = 0$ and $T^2(B/A, M) = 0$ for all M. □

Proposition 1.8. *Suppose that $A = k[x_1, \ldots, x_n]$ and $B = A/I$. Then for any M, there is an exact sequence*

$$0 \longrightarrow T^0(B/k, M) \longrightarrow 0_A(\Omega_{A/k}, M) \longrightarrow 0_B(I/I^2, M) \longrightarrow T^1(B/k, M) \longrightarrow 0$$

and an isomorphism

$$T^2(B/A, M) \simeq T^2(B/k, M).$$

Proof. It follows from the long exact sequence of T^i-functors for $k \to A \to B$ and Propositions A.5, A.6, and A.7. □

Remark 1.9. In this section, we have not made any finiteness assumptions on the rings and modules. It is clear that if A is a Noetherian ring, B is a finitely generated A-algebra, and M is a finitely generated B-module, then $T^i(B/A, M)$ are finitely generated B-modules.

Lemma 1.10. *The construction of the T^i functors is compatible with localization, and thus we may define sheaves $\mathcal{T}^i(X/Y, \mathcal{F})$ for any morphism of schemes $f : X \to Y$ and any sheaf $\mathcal{F}$ of $\mathcal{O}_X$-modules, such that for any open affine $A \subseteq Y$ and any open affine $B \subseteq f^{-1}(A)$, where $\mathcal{F}|_B = \widetilde{M}$, the section $\mathcal{T}^i(X/Y, \mathcal{F})(B) = T^i(B/A, M)$.*

For convenience, denote the modules $T^i(B/A, B)$ and $T^i(B/k, B)$ by $T^i_{B/A}$ and $T^i_{B/k}$ (or T^i_B) if no confusion arises. Furthermore, $T^0_{B/A}$ will be written as the tangent module $T_{B/A}$ of B over A. Similarly for the sheaves $\mathcal{T}^i(X/Y, \mathcal{O}_X)$ and $\mathcal{T}^i(X/k, \mathcal{O}_X)$ we write $\mathcal{T}^i_{X/Y}$ and $\mathcal{T}^i_{X/k}$ (or $\mathcal{T}^i_X$). The sheaf $\mathcal{T}^0_X$ will be written as the tangent sheaf $\mathcal{T}_X$ of X.

Proposition 1.11 (Base change)**.** *Assume that A is Noetherian and B is a finitely generated A-algebra.*

1. *Let $M \in \mathsf{Mod}_B$ and $A \to A'$ be a flat morphism. Let $B' = B \otimes_A A'$ and $M' = M \otimes_B B'$. Then $T^i(B/A, M) \otimes_A A' \simeq T^i(B'/A', M')$ for each i.*
2. *Let $A \to A'$ be an arbitrary ring homomorphism and B is flat over A. Let $B' = B \otimes_A A'$ and $M' \in \mathsf{Mod}_{B'}$. Then $T^i(B/A, M') = T^i(B'/A', M')$ for each i.*

1.2 The infinitesimal Lifting Property

Proposition 1.12 (Infinitesimal Lifting Property)**.** *Let X be a nonsingular affine scheme of finite type over k, let $f : Y \to X$ be a morphism from an affine scheme Y over k, and let $Y \subseteq Y'$ be an infinitesimal thickening of Y. Then the morphism f lifts to a morphism $g : Y' \to X$ such that $g|_Y = f$.*

Proof. Since $Y' = Y$ is affine, we also have Y' is affine. We may transfer the problem to algebra setting. Let $f : A \to B$ be the ring homomorphism over k corresponding to $f : Y \to X$ where $B = B'/I$ with $I^n = 0$ for some n. We want to find a homomorphism $g : A \to B'$ lifting f.

Consider the sequence $B' = B'/I^n \to B'/I^{n-1} \to \cdots \to B'/I^2 \to B'/I$, it will suffice to lift one step at a time. So we may assume that $I^2 = 0$.

Since X is of finite type over k, we may write $A = k[x_1, \ldots, x_n]/J$. Consider any lifting $h : k[x_1, \ldots, x_n] \to B'$ of $k[x_1, \ldots, x_n] \to A \to B$. Then h induces a map $J \to I$, and thus induces $\overline{h} : J/J^2 \to I$ since $I^2 = 0$.

Embed $X \hookrightarrow \mathbb{A}^n_k$ by $P = k[x_1, \ldots, x_n] \to A$. Since X is smooth, we obtain an exact sequence

$$0 \longrightarrow J/J^2 \longrightarrow \Omega^1_{P/k} \otimes_P A \longrightarrow \Omega^1_{A/k} \longrightarrow 0.$$

Notice that those modules correspond to locally free sheaves on X, and thus are projective A-modules. Via the maps h, f, we get a P-module structure on B', and A-module structures on B, I. Applying $0_A(-, I)$ will get another exact sequence

$$0 \longrightarrow 0_A(\Omega^1_{A/k}, I) \longrightarrow 0_P(\Omega^1_{P/k}, I) \longrightarrow 0_A(J/J^2, I) \longrightarrow 0.$$

Let $\theta \in 0_P(\Omega^1_{P/k}, I) = 0_k(P, I)$ be an element whose image is $\overline{h} \in 0_A(J/J^2, I)$. We claim that $h' := h - \theta : P \to B'$ is a ring homomorphism lifting f with $h'(J) = 0$. The first assertion follows by the following easy lemma. □

Lemma 1.13. *Let $B' \to B$ be a surjective homomorphism of k-algebras with kernel I of square zero. Let $R \to B$ be a homomorphism of k-algebras.*

1. *If $f, g : R \to B'$ are two liftings of the map $R \to B$ to B', then $\theta = g - f$ is a k-derivation of R to I.*
2. *Conversely, if $f : R \to B'$ is one lifting, and $\theta : R \to I$ is a derivation, then $g = f + \theta$ is another homomorphism of R to B' lifting the given map $R \to B$.*

In other words, if it is nonempty, the set of liftings $R \to B$ to k-algebra homomorphisms of $R \to B'$ is a principal homogeneous space under the action of $0_k(R, I) = 0_R(\Omega_{R/k}, I)$.

Continuation of the proof of Proposition A.13. Since $h'(J) \subseteq I$, we have $h'(J^2) \subseteq I^2 = 0$. Then h' descends to $P/J^2 \to B'$, and thus $h'|_{J/J^2} = h - h = 0$ by the choice of θ. Hence, h' descends to the desired homomorphism $g : A \to B'$ lifting f. □

Proposition 1.14. *Let X be a scheme of finite type over an algebraically closed field k. Suppose that for every morphism $f : Y \to X$ finite over k, where $Y = R$ is the spectrum of a local Artinian ring R, and for every infinitesimal thickening $Y \subseteq Y'$ with ideal sheaf of square zero, there is a lifting $g : Y' \to X$. Then X is nonsingular.*

Proof. It suffices to show that the local ring $\mathcal{O}_{X,x}$ is a regular local ring for every closed point $x \in X$, so the problem is local and can be reformulated to an algebra problem. Let $(A, \mathfrak{m})$ be a local k-algebra, essentially of finite type over k with residue field k. Assume that for every homomorphism $f : A \to B$, where B is a local Artinian k-algebra, and for every thickening

$$0 \longrightarrow I \longrightarrow B' \longrightarrow B \longrightarrow 0$$

with $I^2 = 0$, there is a lifting $g : A \to B'$. Then we want to show that A is a regular local ring.

Let $a_1, \ldots, a_n \in A$ such that $\{\overline{a}_i\}_{i=1}^n$ form a basis of $\mathfrak{m}/\mathfrak{m}^2$. By the structure theorem of complete local rings, there is a surjective homomorphism $f : P := k[[x_1, \ldots, x_n]] \to \widehat{A}$ defined by $x_i \mapsto a_i$, which induces an isomorphism of $P/\mathfrak{m}_P^2 \to A/\mathfrak{m}^2$.

Consider the thickening $P/\mathfrak{m}_P^{n+1} \to P/\mathfrak{m}_P^n$ and start with the map $A \to A/\mathfrak{m}^2 \simeq P/\mathfrak{m}_P^2$; we can lift to maps $A \to P/\mathfrak{m}_P^n$ for every n by assumption, and thus get a map $g : \widehat{A} \to P$. Then the morphism $P \xrightarrow{f} \widehat{A} \xrightarrow{g} P$ induces an isomorphism on $P/\mathfrak{m}_P^2$.

Let $(A, \mathfrak{m})$ be a local Noetherian k-algebra with residue field k. Let $f \in_k A$ induce an isomorphism on $A/\mathfrak{m}^2$; then f is an isomorphism. If $\Im f \subseteq \mathfrak{m}$, then $A = \Im f + \mathfrak{m}^2 \subseteq \mathfrak{m}$ gives a contradiction. Tensoring with $A/\mathfrak{m}$, we have

$$0 \longrightarrow \ker f/\mathfrak{m} \ker f \longrightarrow A/\mathfrak{m} \xrightarrow{f} A/\mathfrak{m} \longrightarrow 0,$$

and thus $\ker f = 0$ by Nakayama's lemma. By the claim, $g \circ f$ is an automorphism, and hence f is injective. This shows that f is an isomorphism and $\widehat{A}$ is regular, which implies A is also regular. □

Corollary 1.15. *Let X be a nonsingular affine scheme over k. Let A be a local Artinian ring over k, and let X' be a scheme flat over A such that $X' \times_A k$ is isomorphic to X. Then X' is isomorphic to the trivial deformation $X \times_k A$ of X over A.*

Proof. Apply Proposition A.13 to ${}_X : X \to X$ and the infinitesimal thickening $i : X \to X'$; there exists a lifting $p : X' \to X$ such that $p \circ i =_X$. This defines a map $f : X' \to X \times_k A$ of flat schemes over A, which induces the identity map on the closed fiber X. By the following claim, f is an isomorphism.

Let A be a local Artinian k-algebra. Let $f : X_1 \to X_2$ be an A-morphism between two finite type flat schemes over A, which induces an isomorphism of closed fibers $f \otimes_A k : X_1 \times_A k \to X_2 \times_A k$. Then f is an isomorphism. Since $X_i \times_A k \to X_i$ is an isomorphism of topological spaces, we only need to show that f induces an isomorphism on each stalk of the structure sheaf. We may check this locally. Let $f : B_2 \to B_1$ be the ring homomorphism corresponding to open affine neighborhoods of x and $f(x)$ respectively. Let $\mathfrak{p} \in B_1$ correspond to x and $\mathfrak{q} = f^{-1}(\mathfrak{p})$. Then f induces a morphism $g : (B_2)_{\mathfrak{q}} \to (B_1)_{\mathfrak{p}}$, which induces an isomorphism $g \otimes_A A/\mathfrak{m}_A$ by assumption. By the long exact sequence of Tor, we have $g = 0$. Since B_i are flat over A, we have

$$0 =_1^A ((B_1)_{\mathfrak{p}}, A/\mathfrak{m}_A) \longrightarrow \ker g \otimes_A A/\mathfrak{m}_A \longrightarrow (B_2)_q \otimes_A A/\mathfrak{m}_A \longrightarrow (B_1)_p \otimes_A A/\mathfrak{m}_A \longrightarrow 0,$$

and thus $\ker g = 0$ by Nakayama's lemma. □

Next, we investigate the relation between nonsingularity and the T^i functors.

Theorem 1.16. *Let $X = B$ be an affine scheme over an algebraically closed field k. Then X is nonsingular if and only if $T^1(B/k, M) = 0$ for all $M \in \mathsf{Mod}_B$. Furthermore, if X is nonsingular, then $T^2(B/k, M) = 0$ for all M.*

Proof. Write B as a quotient of a polynomial ring $A = k[x_1, \ldots, x_n]$ over k. Then X is nonsingular if and only if the conormal sequence

$$0 \longrightarrow I/I^2 \longrightarrow \Omega_{A/k} \otimes_A B \longrightarrow \Omega_{B/k} \longrightarrow 0 \tag{12}$$

is exact and $\Omega_{B/k}$ is locally free, i.e., a projective B-module. Since $\Omega_{A/k}$ is a free A-module, we conclude that X is nonsingular if and only if (12) is split exact. From the four-term exact sequence in Proposition A.9

$$0 \longrightarrow T^0(B/k, M) \longrightarrow 0(\Omega_{A/k}, M) \longrightarrow 0(I/I^2, M) \longrightarrow T^1(B/k, M) \longrightarrow 0,$$

$T^1(B/k, M) = 0$ for all $M \in \mathsf{Mod}_B$ if and only if the map

$$0(\Omega_{A/k}, M) \longrightarrow 0(I/I^2, M)$$

is surjective for all M, which is equivalent to split exactness by considering $M = I/I^2$.

Again, by Proposition A.9 we have $T^2(B/k, M) = T^2(B/A, M)$. Since X is smooth, the localizing ideal I_x at $x \in X$ is generated by a regular sequence in the regular local ring A_x. By Corollary A.8, $T^2(B_x/A_x, M) = 0$ for all $M \in \mathsf{Mod}_{B_x}$. By Lemma A.11, we conclude that $T^2(B/A, M) = 0$ for all $M \in \mathsf{Mod}_B$. □

Theorem 1.17. *A morphism of finite type $f : X \to Y$ of Noetherian schemes is smooth if and only if it is flat, and $\mathcal{T}^1(X/Y, \mathcal{F}) = 0$ for all coherent sheaves $\mathcal{F} \in \mathcal{C}oh(X)$. Furthermore, if f is smooth, then $\mathcal{T}^2(X/Y, \mathcal{F}) = 0$ for all $\mathcal{F}$.*

Proof. The problem is local, so we may assume that $X = B$ and $Y = A$ are affine and f is given by a ring homomorphism $A \to B$.

First suppose B is flat over A and $T^1(B/A, M) = 0$ for all $M \in \mathsf{Mod}_B$. Let $y \in Y$ correspond to $\mathfrak{p} \in A$ and let $k = k(y)$ be its residue field. Let $A' = A/\mathfrak{p}$ and $B' = B \otimes_A A' = B/\mathfrak{p}B$; then for any $M \in \mathsf{Mod}_{B'}$ we obtain $T^1(B'/A', M) = T^1(B/A, M) = 0$ by base change A.12. Write $B' = A'[x_1, \ldots, x_n]/I$; then $T^1(B'/A', I/I^2) = 0$.

Now consider the flat base extension from A' to $\overline{k} = \overline{(A')}$. By base change A.12, we have

$$T^1(B' \otimes_{A'} \overline{k}/\overline{k}, (I/I^2) \otimes \overline{k}) = 0.$$

Since $\overline{k}$ is flat over A', we have $B' \otimes \overline{k} = \overline{k}[x_1, \ldots, x_n]/\overline{I}$ and $(I/I^2) \otimes \overline{k} = \overline{I}/\overline{I}^2$, where $\overline{I} = I \otimes \overline{k}$. Then the proof of A.17 shows that $B' \otimes \overline{k}$ is nonsingular over $\overline{k}$. Hence, the geometric fibers of f are nonsingular, and thus f is smooth.

Conversely, suppose that B is smooth over A.

$T^1(B/A, B/\mathfrak{m}) = 0$ for all maximal ideals $\mathfrak{m}$ of B. Let $\mathfrak{m}$ correspond to the point $x \in B$, let $f(x) = y$, and let $k = k(y)$. Apply base change on $B/\mathfrak{m} \in \mathsf{Mod}_{B \otimes_A k}$, we obtain

$$T^1(B/A, B/\mathfrak{m}) \otimes_k \overline{k} = T^1(B \otimes_A k/k, B/\mathfrak{m}) \otimes_k \overline{k} = T^1(B \otimes_A \overline{k}/\overline{k}, (B/\mathfrak{m}) \otimes_k \overline{k}) = 0$$

since the geometric fibers are nonsingular. Since $k \to \overline{k}$ is faithfully flat, we conclude that $T^1(B/A, B/\mathfrak{m}) = 0$.

Lemma 1.18 (Devissage)**.** *Let B be a Noetherian ring, and let F be a semi-exact (that is, F applied to a short exact sequence is exact in the middle) additive functor from finitely generated B-modules to itself. Assume that $F(B/\mathfrak{m}) = 0$ for every maximal ideal $\mathfrak{m}$ of B. Then $F(M) = 0$ for all finitely generated B-modules.*

Let $0 = M_n \subseteq \cdots \subseteq M_1 \subseteq M_0 = M$ be a composition series such that $M_i/M_{i+1} \simeq B/\mathfrak{p}_i$ for some $\mathfrak{p}_i \in B$. By semi-exactness, it suffices to prove the case for $M = B/\mathfrak{p}$.

We induct on $\dim M$. If $\dim M = 0$, then $M = B/\mathfrak{m}$ for some maximal ideal, and thus $F(M) = 0$ by hypothesis. For the general case, let $M = B/\mathfrak{p}$. For any maximal ideal $\mathfrak{m} \supseteq \mathfrak{p}$, pick an element $t \in \mathfrak{m} \backslash \mathfrak{p}$. Then t is a non-zero divisor for M and $M' := (M \xrightarrow{t} M)$ has support of dimension $< \dim M$. By the

induction hypothesis, $F(M) \xrightarrow{t} F(M)$ is surjective. By Nakayama's lemma, $F(M)_{\mathfrak{m}} = 0$. Hence, $F(M) = 0$.

Since $T^1(B/A, -)$ satisfies the condition in the Devissage lemma, we have $T^1(B/A, M) = 0$ for all finitely generated B-modules, and hence for all B-modules, because T^i commutes with direct sums. The same argument shows also that $T^2(B/A, M) = 0$ for all M. □

Theorem 1.19. *Let A be a regular local k-algebra with residue field $k = \overline{k}$, and $B = A/I$. Then B is a local complete intersection in A if and only if $T^2(B/k, M) = 0$ for all $M \in \mathsf{Mod}_B$.*

Proof. Since A is regular, we have $T^1(A/k, M) = 0$ for $i = 1, 2$ and all M by Theorem A.17. From the nine-term exact sequence, we have $T^2(B/k, M) = T^2(B/A, M) = 0$ by Corollary A.8.

Conversely, suppose that $T^2(B/k, M) = 0$ for all M. As above, this implies $T^2(B/A, M) = 0$ for all M. To compute this group, we may take $R = A$, $I = I$, and let F map to a minimal set of generators $(a_1, \ldots, a_s)$ of I, with kernel Q. Then the hypothesis $T^2(B/A, M) = 0$ for all M implies that

$$0(F/IF, M) \to 0(Q/F_0, M)$$

is surjective for all M. In particular for $M = Q/F_0$ and the map $d_2 : Q/F_0 \to F/IF$, there exists $p : F/IF \to Q/F_0$ such that $p \circ d_2 =_{Q/F_0}$. By Nakayama's lemma, $Q \subseteq \mathfrak{m}_A F$. Then the identity map $p \circ d_2$ sends Q/F_0 into $\mathfrak{m}_A(Q/F_0)$, and thus $Q/F_0 = 0$ by Nakayama's lemma. But Q/F_0 is actually the first homology group of the Koszul complex $K_\bullet(a_1, \ldots, a_s)$ over A, and the vanishing of this group is equivalent to $a_1, \ldots, a_s$ being a regular sequence. □

Remark 1.20. If we define a relative local complete intersection morphism $f : X \to Y$ to be a flat morphism whose geometric fibers are local complete intersection schemes, then f is a relative local complete intersection morphism if and only if $\mathcal{T}^i(X/Y, \mathcal{F}) = 0$ for all coherent sheaves $\mathcal{F}$ on X.

1.3 Deformations of Rings

Proposition 1.21 (Local criterion of flatness)**.** *Let $A' \to A$ be a surjective homomorphism of Noetherian rings whose kernel J is square zero. Then an A'-module M' is flat over A' if and only if*

1. *$M = M' \otimes_{A'} A$ is flat over A, and*
2. *the natural map $M \otimes_A J \to M'$ is injective.*

References

[1] Hartshorne, R. *Deformation Theory.* Graduate Texts in Mathematics, 257. Springer, 2010.